

BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `~/BEE 4750 /HW 2/hw2-regina`
Installed Fontconfig_jll      v2.17.1+0
Installed Libmount_jll        v2.41.1+0
Installed WorkerUtilities      v1.6.1
Installed JpegTurbo_jll        v3.1.3+0
Installed Accessors            v0.1.42
Installed Preferences          v1.5.0
Installed Roots                v2.2.10
Installed DataFrames           v1.8.0
Installed WeakRefStrings       v1.4.2
Installed DataStructures       v0.19.1
Installed ConstructionBase      v1.6.0
Installed Expat_jll            v2.7.1+0
Installed Ghostscript_jll      v9.55.1+0
Installed StatsBase            v0.34.6
Installed CompositionsBase     v0.1.2
Installed CommonSolve          v0.2.4
Installed InverseFunctions      v0.1.17
Installed Plots                 v1.40.19
Installed PrettyTables         v3.0.8
Installed Libuuid_jll          v2.41.1+0
Installed OpenSSL_jll          v3.5.2+0
Installed Latexify              v0.16.10
Installed Glib_jll             v2.86.0+0
Installed FilePathsBase        v0.9.24
Installed SortingAlgorithms    v1.2.2
Installed CSV                   v0.10.15
Precompiling project...
 538.4 ms   WorkerUtilities
 480.1 ms   CompositionsBase
 482.0 ms   InverseFunctions
 479.6 ms   ConstructionBase
 481.7 ms   CommonSolve
 596.5 ms   Preferences
 693.6 ms   FilePathsBase
 687.2 ms   LogExpFunctions → LogExpFunctionsInverseFunctionsExt
 693.9 ms   InverseFunctions → InverseFunctionsDatesExt
 733.6 ms   Unitful → InverseFunctionsUnitfulExt
 811.3 ms   InverseFunctions → InverseFunctionsTestExt
1313.9 ms   DataStructures
 870.2 ms   CompositionsBase → CompositionsBaseInverseFunctionsExt
 495.7 ms   ConstructionBase → ConstructionBaseLinearAlgebraExt
 436.6 ms   Unitful → ConstructionBaseUnitfulExt

```

623.0 ms	PrecompileTools
700.0 ms	JLLWrappers
940.5 ms	FilePathsBase → FilePathsBaseMmapExt
986.4 ms	SortingAlgorithms
1417.3 ms	FilePathsBase → FilePathsBaseTestExt
650.5 ms	Graphite2_jll
973.2 ms	OpenSSL_jll
1831.8 ms	RecipesBase
1882.8 ms	StringManipulation
485.8 ms	Libmount_jll
466.9 ms	EpollShim_jll
424.7 ms	LLVMOpenMP_jll
448.6 ms	Bzip2_jll
401.3 ms	Xorg_libICE_jll
399.4 ms	Xorg_libXau_jll
2746.1 ms	Accessors
464.2 ms	libpng_jll
405.2 ms	libfdk_aac_jll
401.4 ms	LAME_jll
391.6 ms	LERC_jll
426.3 ms	fzf_jll
368.2 ms	mtdev_jll
383.4 ms	Ogg_jll
441.7 ms	JpegTurbo_jll
475.9 ms	XZ_jll
389.6 ms	Xorg_libXdmcp_jll
422.8 ms	x265_jll
423.2 ms	x264_jll
447.9 ms	libaom_jll
450.7 ms	Zstd_jll
411.8 ms	Expat_jll
3469.1 ms	ColorSchemes
459.0 ms	Opus_jll
472.6 ms	LZO_jll
462.9 ms	Xorg_xtrans_jll
445.8 ms	libevdev_jll
532.0 ms	Libiconv_jll
494.1 ms	Libffi_jll
425.2 ms	Libuuid_jll
438.7 ms	eudev_jll
469.5 ms	FriBidi_jll
344.9 ms	Xorg_libSM_jll
875.1 ms	Pixman_jll

1060.9 ms	FreeType2_jll
1136.9 ms	Accessors → LinearAlgebraExt
1984.1 ms	StatsBase
773.6 ms	Accessors → TestExt
648.0 ms	Accessors → UnitfulExt
479.7 ms	Xorg_libxcb_jll
882.2 ms	libvorbis_jll
959.4 ms	JLFzf
3154.8 ms	OpenSSL
448.6 ms	DBus_jll
1768.2 ms	Ghostscript_jll
475.9 ms	Wayland_jll
608.9 ms	libinput_jll
1192.6 ms	GettextRuntime_jll
2053.0 ms	Libtiff_jll
8805.7 ms	Parsers
360.1 ms	Xorg_xcb_util_jll
335.2 ms	Xorg_libX11_jll
2223.8 ms	Fontconfig_jll
2630.7 ms	Roots
2079.4 ms	Glib_jll
1069.2 ms	InlineStrings → ParsersExt
368.3 ms	Xorg_xcb_util_image_jll
337.3 ms	Xorg_xcb_util_keysyms_jll
378.4 ms	Xorg_xcb_util_renderutil_jll
481.6 ms	Xorg_xcb_util_wm_jll
1131.6 ms	JSON
681.6 ms	Xorg_libXrender_jll
3602.8 ms	Latexify
692.7 ms	Xorg_libXext_jll
636.0 ms	Xorg_libXfixes_jll
556.0 ms	Xorg_libxkbfile_jll
584.5 ms	Xorg_xcb_util_cursor_jll
926.9 ms	Roots → RootsUnitfulExt
907.5 ms	WeakRefStrings
427.3 ms	Xorg_libXinerama_jll
443.8 ms	Xorg_libXrandr_jll
790.3 ms	Latexify → SparseArraysExt
437.9 ms	Libglvnd_jll
7422.9 ms	PlotUtils
1317.6 ms	UnitfulLatexify
428.5 ms	Xorg_libXcursor_jll
409.0 ms	Xorg_libXi_jll

400.6 ms	Xorg_xkbcomp_jll
1166.8 ms	Xorg_xkeyboard_config_jll
371.8 ms	xkbcommon_jll
557.2 ms	Vulkan_Loader_jll
2378.4 ms	PlotThemes
3760.9 ms	RecipesPipeline
4986.4 ms	Cairo_jll
1403.2 ms	HarfBuzz_jll
1220.8 ms	libass_jll
1243.5 ms	Pango_jll
4851.7 ms	Qt6Base_jll
403.9 ms	libdecor_jll
432.3 ms	GLFW_jll
13871.3 ms	HTTP
1057.0 ms	Qt6ShaderTools_jll
2684.2 ms	FFMPEG_jll
1574.2 ms	Qt6Declarative_jll
358.1 ms	Qt6Wayland_jll
747.7 ms	FFMPEG
1232.9 ms	GR_jll
21617.3 ms	PrettyTables
11840.1 ms	CSV
2247.9 ms	GR
24016.3 ms	DataFrames
1459.1 ms	Latexify → DataFramesExt
33918.9 ms	Plots
2830.6 ms	Plots → UnitfulExt

128 dependencies successfully precompiled in 67 seconds. 91 already precompiled.

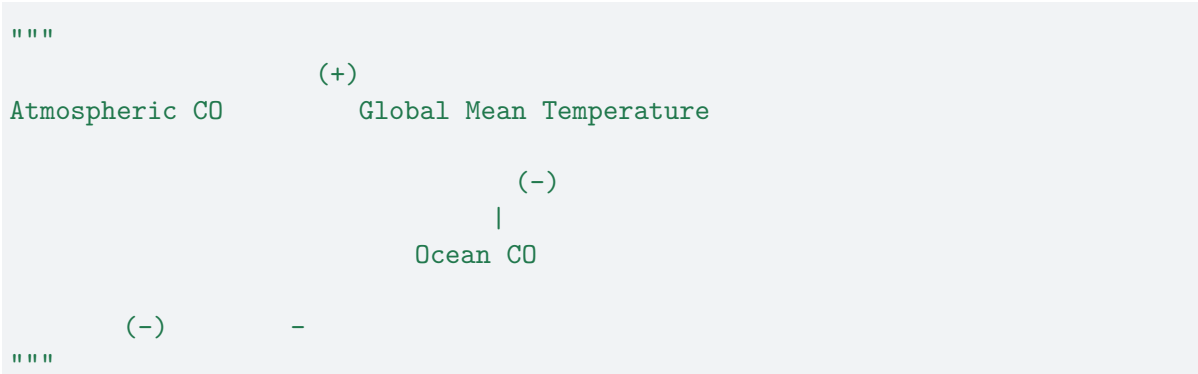
```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions

between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?



```

"      (+)\nAtmospheric CO      Global Mean Temperature\n

```

Problem 1 SOLUTION:

Henry's Law says that solubility of CO₂ in seawater decreases as temperature increases. In other words, warm water holds less CO₂. Because of this, the arrow pointing from global mean temperature to the concentration of CO₂ in the ocean is a negative (-) sign and this is a dampening relationship. As temperature increases, the ocean's water warms and CO₂ concentrations decrease.

When concentrations of CO₂ in the ocean increase, the concentration of CO₂ in the atmosphere decrease since the ocean is acting as a carbon sink and taking out CO₂ from the atmosphere. Because of this, the sign is negative (-) and this is a dampening relationship.

Finally, when CO₂ concentrations in the atmosphere increases, the greenhouse effect is magnified since the CO₂ is now absorbing heat and re-emitting it. As a result, the global mean temperature to increase. This is an amplifying relationship and the arrow's sign is positive (+).

This suggests that the overall feedback between temperature and the ocean carbon cycle is an amplifying (or positive) feedback because an increase in atmospheric CO₂, increases temperature, which creates less CO₂ in the ocean, which means less CO₂ is being absorbed by the ocean and there is more of it in the atmosphere, which finally leads to even more warming.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

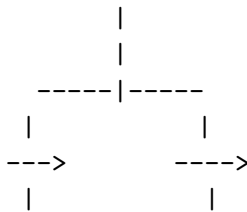
A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d .

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

2.1 Solution:



Two boxes or control volumes are needed for this because there are two input streams, one before $X = 15 \text{ km}$ and one after. These streams are causing a change in flow rate of the river so the control volumes will help model and understand those changes.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

References

List any external references consulted, including classmates.